

A communication subspace relays context-dependent actions from human prefrontal to motor cortex

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Adaptive behavior relies on the ability to translate abstract rules and goals into actions suited to the current context. Neural population activity in the prefrontal cortex (PFC) has been proposed to support such flexible computations through high-dimensional dynamics, whereas activity in the primary motor cortex (M1) is related more directly to movement execution. How contextual representations in PFC are transformed into ensuing action plans within M1 remains unknown. Previous work suggests that low-dimensional coding subspaces might organize interareal communication, but direct evidence for such population-level communication mechanisms in humans is lacking. Here we use intracranial recordings from human PFC and M1 to identify a communication subspace embedded within high-dimensional PFC activity, that selectively relays behaviorally relevant information at the single-trial level. Activity in this subspace predicts context-dependent action more strongly than either region, revealing a fundamental coding principle by which coordinated interareal population dynamics filter and relay predictive information to guide context-dependent actions.

Context-dependent behavior is a hallmark of human cognitive flexibility, allowing actions to be selected based on current rules, task demands, long-term goals and past experiences^{1–3}. At the core of this adaptability lies predictive processing, which integrates exogenous and endogenous priors to anticipate relevant outcomes, shape sensory processing and optimize action selection^{4,5}. Although distributed networks in the PFC provide the structural basis for these neural computations^{6–8}, the functional mechanisms governing their adaptive processing remain poorly understood. A central challenge in identifying the neural correlates of such complex human behavior is the high-dimensional nature of neurophysiological activity, particularly in association areas such as the PFC^{2,9–12}. This high-dimensionality reflects directly the complex, multiplexed processes that continuously integrate sensory information to drive

context-dependent action execution^{1,3,13}. Mechanisms such as recurrent connections^{3,14}, interareal connectivity^{6,15} and mixed selectivity^{11,12,16}, among other presumed mechanisms, support this high-dimensional nature and render it difficult to isolate behaviorally relevant signatures from population activity. In contrast to complex PFC dynamics, population activity in motor cortex (M1) often unfolds along well-structured trajectories^{17–20}. The PFC–M1 network is thought to support adaptive behavior by routing relevant information dynamically from PFC to M1 (refs. [15,21,22](#)). Consistent with the presumed directional communication pathway, previous studies demonstrated that sensory- and context-related activity emerges first in the PFC and then in M1 (refs. [15,21](#)). However, it remains puzzling how multifaceted representations of contextual information in PFC are transformed into ensuing motor plans.

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Growing evidence suggests that interareal communication is structured through low-dimensional coding subspaces^{23–26}. This concept aligns with the population doctrine, which posits that cell assemblies and their geometric configuration, not individual neurons, constitute the central computational unit of the brain^{27–29}. However, direct experimental evidence for such communication subspaces (CSs) remains sparse^{24–26,30} and is confined largely to studies in nonhuman primates and mice³¹. Although coding subspaces have been described in humans at the level of aggregated single unit^{32,33} and field potentials^{34–36}, it remains unclear whether such low-dimensional subspaces also support PFC–M1 communication in humans and, critically, whether they have a role in higher-order cognition and goal-directed behavior.

Here we determined how context-dependent action information is encoded in the PFC and transformed into motor output to support efficient visual target detection. We hypothesized that PFC and M1 share a joint subspace, analogous to previously studied CSs observed between early and intermediate visual system. Specifically, we reasoned that this subspace might act as a filter for selectively relaying contextually driven action signals from PFC to M1, thereby channeling PFC activity efficiently to downstream motor circuits. If true, then neural activity within this CS should correlate better with context-dependent action than activity within either region alone. The identification of such subspaces requires sufficient spatiotemporal sampling of the underlying neural population. As previous studies relied on aggregated unit firing, we leveraged high-frequency activity (HFA)—a well-established proxy for coordinated population activity³⁷—recorded from densely sampled subdural grid electrodes (electrocorticography, ECoG) covering large portions of PFC and M1 (refs. 38,39). Our findings uncover a shared PFC–M1 CS in which coordinated low-dimensional activity structures information flow from PFC to M1 to enable context-guided target detection, thereby establishing a fundamental principle of neural communication at the population level that supports efficient information transfer in support of goal-directed actions.

Results

To test whether a low-dimensional PFC–M1 CS acts as a selective routing mechanism for filtering context-dependent action information within PFC to M1, we recorded subdural ECoG signals using grid electrodes from 12 pharmacoresistant epilepsy patients (age 40.91 ± 12.32 years, mean $\pm$ s.d.; four female, one unknown) who performed a context-guided target detection task^{40,41}. Participants viewed a continuous stream of triangles with varying orientations and were instructed to respond as quickly as possible when a downward facing triangle (target stimulus (T)) appeared (see Supplementary Table 1 for additional details about the task). Targets were either presented at random or preceded by a fully predictive sequence (100% validity, left-, up- and then right-ward facing triangles; Fig. 1a). Reaction times (RTs) were significantly faster for predicted targets compared to random targets ($P = 0.0333$; $t_{11} = -2.43$, $d = -0.70$; Fig. 1b), indicating that participants relied on the predictive temporal context for effective target detection. To investigate how the prefrontal motor network encodes and transforms this predictive information to guide behavior, we analyzed the simultaneously recorded ECoG signals (PFC: 74 ± 29 contacts, mean $\pm$ s.e.m., M1: 26 ± 7 ; Fig. 1c), using the bipolar referencing scheme. We leveraged specifically ECoG recordings, which provide dense spatial sampling of both regions, to test whether behaviorally relevant information is relayed through structured low-dimensional codes. We hypothesized that a PFC–M1 CS could serve as a filter mechanism to funnel behaviorally relevant information nested in high-dimensional prefrontal dynamics to the lower dimensional motor cortex (Fig. 1d).

Target-related HFA in human PFC and motor cortex

To uncover the neural correlates underlying the behavioral benefit afforded by predicted targets, we analyzed the simultaneously recorded neural activity from the human PFC and M1 for 11 participants (one

patient had to be excluded for excessive high-frequency artifacts). We employed a bipolar referencing scheme along all cardinal and diagonal directions of the ECoG grid (typical dimensions: 8×8 electrodes) but included only unique pairs that could be assigned unambiguously to either the PFC or M1. We quantified HFA as a proxy for coordinated population activity, time-locked to target presentation ($t = 0$). Although we observed a clear increase in target-related activity, there were no significant differences in the overall activity level between predicted and random targets in PFC and M1 (Fig. 2a; cluster-based permutation test for paired samples across timepoints, corrected for multiple comparisons, no clusters were found at $p < 0.05$). However, we did observe that HFA peaks for the predictive trials occurred earlier in the PFC compared to in the random trials (Wilcoxon signed-rank test; $P = 0.0003$) but this effect was not seen in M1 (Wilcoxon signed-rank test; $P = 0.3468$; Fig. 2a). To test whether these context-dependent temporal dynamics were linked to behavior, we examined whether peak latency difference of context-specific target-related activity correlated with the individual behavioral benefit of the predictive relative to the random context (Fig. 2a) but observed no significant correlations in either region (Spearman rank-order correlation; PFC: $r = 0.39$, $P = 0.2345$; M1: $r = 0.35$, $P = 0.2847$).

High-dimensional neural dynamics in human PFC

Although clear target-related activity was evident in both PFC and M1 (Fig. 2a), neither overall activity nor peak latency differences correlated significantly with the prediction effect (Fig. 1b). We hypothesized that this absence was the direct consequence of the high-dimensional nature of prefrontal neural population activity that reflects simultaneously both task-relevant and task-irrelevant features, potentially masking behaviorally relevant activity. Consequently, we considered that context-specific action information might reside only in a subspace of the overall neural population activity (Fig. 1d). To test this hypothesis, we first estimated the intrinsic dimensionality of neural activity in human PFC and M1 using factor analysis⁴² (Methods). At the single-participant level, more factor analysis components were required to explain the full variance in PFC as compared to M1, indicating higher complexity of PFC population activity (Fig. 2b, left). The difference was also evident at the group level, with significantly higher dimensionality in PFC compared to M1 ($P = 0.0174$, $t_{10} = 2.85$, $d = 0.85$; Fig. 2b, middle), a pattern that was consistent across individual participants (Fig. 2b, right). To confirm that this effect was not specific to the choice of dimensionality reduction method, we repeated the analysis using participation ratio—a common alternative measure—which yielded the same result (Supplementary Fig. 1). In summary, this set of findings demonstrates that human PFC population activity is more complex (that is, higher dimensional) than the population activity recorded from M1.

Identification of a low-dimensional CS within the human PFC during context-guided behavior

Having established the high-dimensional nature of human prefrontal population activity, we then investigated whether a low-dimensional subspace within PFC could selectively capture the action information that is relayed downstream to M1. We first characterized PFC–M1 interactions by assessing how well variability in the M1 population activity could be predicted from the PFC population activity, using reduced rank regression (RRR)⁴³ (Fig. 3a and Methods). We then identified the specific dimensions mediating this communication between the two cortical regions^{23,24}, which we termed the PFC–M1 CS. If the communication between the two regions occurs within a subset of the PFC population activity, then a low-dimensional subspace should achieve the same predictive performance as the full regression model (Methods). To test this hypothesis, we constructed a CS for each participant independently. To minimize bias and account for variability in electrode coverage and trial count across participants, we repeatedly

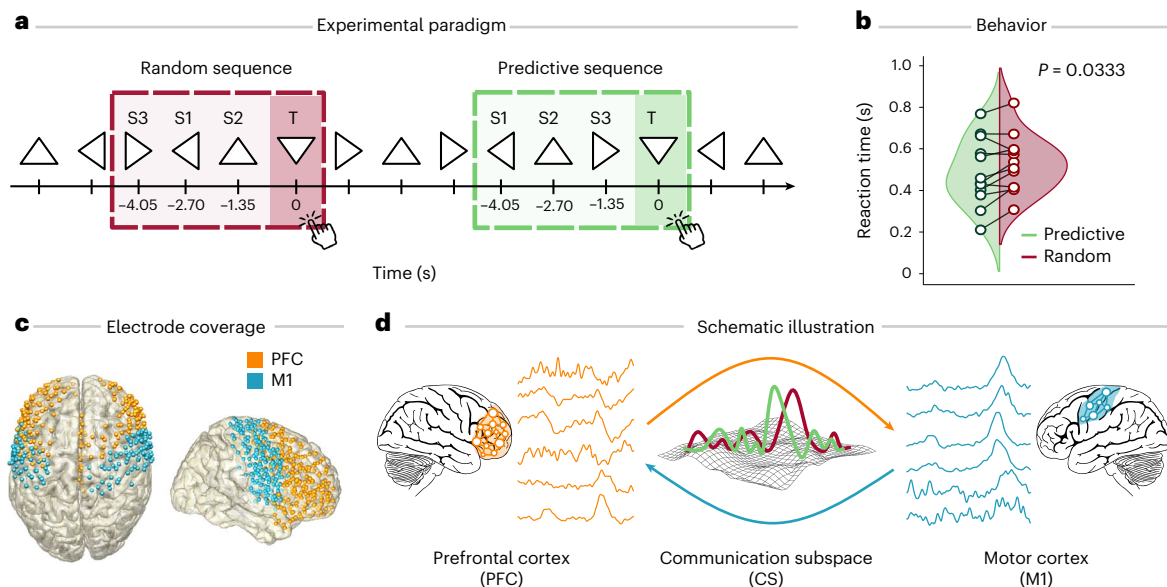


Fig. 1 | Experimental design, electrode placement, behavior and key hypothesis. **a**, Experimental design: participants performed a visual target detection task that featured random and predicted targets (150-ms stimulus presentation, 1.2-s interstimulus intervals). Random targets (red box) were preceded by stimuli that were not presented in a specific order, whereas predicted targets (green box) were signaled by a preceding sequence (clockwise rotating triangles: S1, S2, S3). Targets (T) were presented throughout the presentation blocks and required a button press in response to the target presentation. **b**, Behavioral difference in RTs: group-level distributions of RTs for predictive (green) and random (red) targets, with faster RTs for predictive context ($n = 12$; $P = 0.0333$, $t_{11} = -2.43$, $d = -0.70$; two-sided paired t -test).

c, Electrode coverage in PFC and M1: subdural grid bipolar contacts across 12 participants normalized to an MNI template brain. Left: top view; right: lateral view with electrodes from the left hemisphere mirrored and superimposed onto the right hemisphere for visualization. Electrode contacts are shown over the PFC (orange, 885 contacts) and motor cortex (blue: M1; 316 contacts). **d**, Schematic illustration of the key hypothesis: context-relevant information (green: predictive; red: random) may be encoded within a subspace of high-dimensional PFC population activity. We hypothesized that high-dimensional activity in PFC is projected into a low-dimensional CS that allows the selective relay of only task-relevant information to M1, to effectively guide action execution.

subsampled a fixed number of channels ($n = 15$; Methods) from both regions and averaged trials within each context condition (predictive or random). As hypothesized, the identified PFC–M1 subspace captured the interactions between the human PFC and M1 within a subset of dimensions (PFC–M1, purple) of the total neural population space (full model, gray; Wilcoxon signed-rank test; $P = 0.0010$; Fig. 3b, left). To quantify how the interaction between PFC and M1 differs from interactions within PFC, we also constructed a PFC–PFC subspace, where PFC was split randomly into two subsets, with one predicting the other. For a representative single participant (Fig. 3b, right), six dimensions sufficed to match the predictive performance (R^2) of the full regression model ($R^2 = 0.58 \pm 0.16$; computed with 15 dimensions in this case) for PFC–M1 communication. By contrast, the PFC–PFC subspace required eight dimensions to achieve similar performance. Adding more dimensions to the reduced model did not improve the predictive performance (R^2), indicating that communication both within and across cortical regions occur through a low-dimensional subspace. At the group level, significantly fewer dimensions were needed to capture the PFC–M1 interactions as compared to the interactions within PFC ($P = 0.0010$, $t_{10} = -5.95$, $d = -1.79$; Fig. 3c, left), revealing that the PFC–M1 subspace is lower dimensional than subspaces capturing interactions within PFC. Moreover, the PFC–PFC subspace consistently exhibited higher predictive performance than the PFC–M1 subspace ($P = 0.0023$, $t_{10} = -4.05$, $d = -1.22$; Fig. 3c, right). These results highlight that within-PFC interactions are more complex than the interactions between PFC and M1. To ensure that the lower dimensionality of the PFC–M1 subspace was not inherited trivially from the lower dimensionality of M1, we performed a control analysis in which the same M1 target population was used to compare PFC–M1 interactions with matched within-M1 interactions (Methods). Because both models predicted the same M1 activity, any constraint imposed by M1 affected them equally. Under these matched conditions, the PFC–M1 subspace

remained significantly lower dimensional than the within-M1 subspace ($P = 0.0078$, Wilcoxon signed-rank test; Supplementary Fig. 2), confirming that the low dimensionality of the PFC–M1 CS reflects a selective projection from PFC rather than a constraint imposed by the downstream M1 population. Critically, our findings establish the existence of a low-dimensional CS between PFC and M1 in humans, consisting of a small subset of the overall PFC population activity.

Low-dimensional CS tracks predictive information

After identifying the PFC–M1 CS defined solely on neural population activity patterns, we examined whether this subspace plays a functional role in mediating the context-dependent advantage in behavioral performance. We first extracted the latent PFC activity pattern within this CS by projecting the PFC HFA onto its first dimension (Fig. 4a and Methods). We focused on the first dimension because it consistently captured the largest amount of variance in the CS across participants (Supplementary Fig. 3). This projection reduced the dimensionality of the PFC population activity, while preserving the temporal structure of the neural dynamics that is most predictive of M1 activity⁴⁴. We then assessed the relationship between this low-dimensional representation of PFC and target detection performance. We observed a clear peak in the latent dynamics of the PFC–M1 CS after target onset, suggesting that neural activity within the CS is modulated by target-related processing (Fig. 4a; example participants). Notably, these peaks generally occurred earlier in predictive compared to random contexts for participants who showed a strong behavioral effect (Fig. 4a). To quantify the association between behavior and the latent dynamics of PFC within the PFC–M1 CS, we correlated the peak latency differences between predictive and random peaks of this latent activity with the context-dependent differences in mean RTs across participants. We observed a statistically significant positive correlation ($r_{CS} = 0.79$; $P = 0.0037$, Spearman rank-order correlation; Fig. 4b, left), indicating

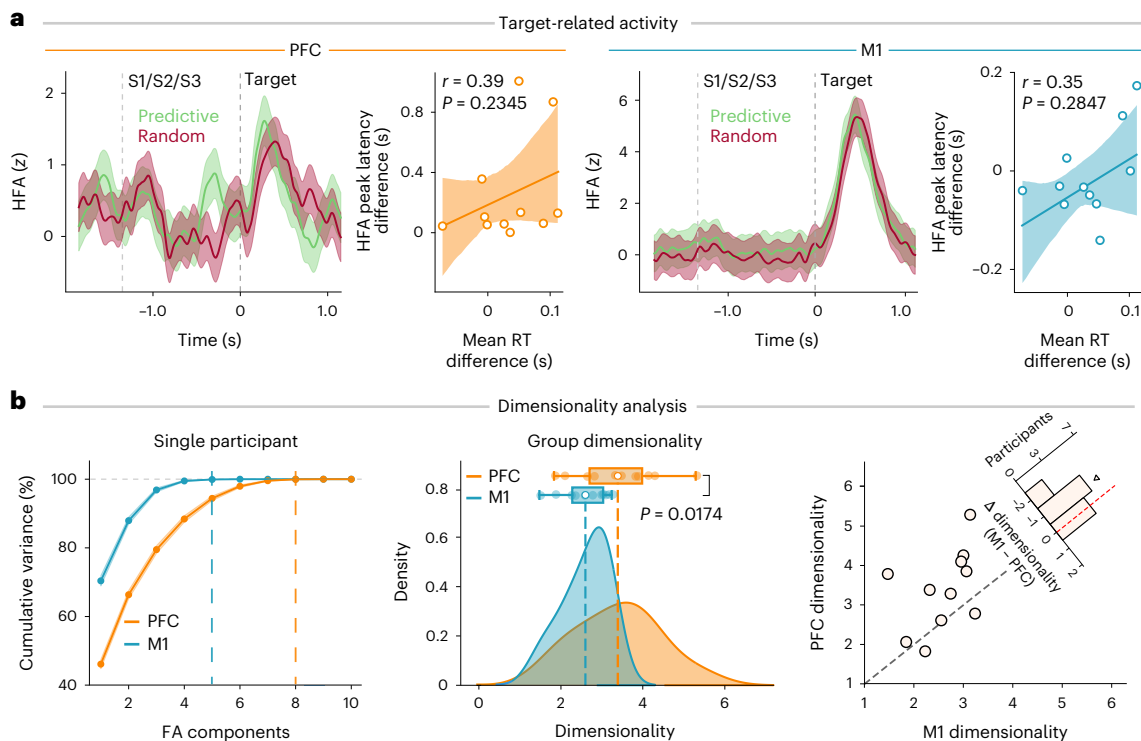


Fig. 2 | HFA responses and dimensionality of target-related neural activity in PFC and M1. **a**, Target-related HFA activity: z-scored HFA timecourses during target presentation are shown for PFC and M1 for predictive (green) and random (red) context. In both regions, HFA did not show any significant differences between context-conditions (shaded areas: s.e.m.). Group-level correlation between HFA peak latency differences and mean RT differences (shaded areas: 95% CI) demonstrate no significant relationship for either PFC ($r = 0.39$, $P = 0.2345$; two-sided Spearman rank correlation) or M1 ($r = 0.35$, $P = 0.2847$; two-sided Spearman rank correlation). **b**, Dimensionality analysis of neural population activity. Left: percentage cumulative variance explained by different dimensions in a representative participant using factor analysis. Cumulative variance of the first ten factors is visualized for PFC (orange) and M1 (blue). M1 achieves 100% explained variance with fewer factors needed as compared to PFC, indicating lower dimensionality of the M1 population activity. Dashed lines: point where

100% cumulative variance is reached. Shaded areas: ± 1 s.e.m. around the mean across trials. Middle: group-level distribution of dimensionality, with significantly higher PFC dimensionality as compared to M1 ($n = 11$; $P = 0.0174$, $t_{10} = 2.85$, $d = 0.85$; two-sided paired t -test). Boxes indicate the interquartile range (25–75%) and whiskers extend to $1.5 \times$ interquartile range. Circles: optimal dimensionality (computed across trials; Methods) for individual participants; white circles (and dashed lines): group-level mean dimensionality for each region. Right: trial-averaged dimensionality across individual participants in PFC and M1 signal greater complexity in neural population activity in PFC as compared to M1. Each circle represents a single participant, with the inset showing the distribution of difference in dimensionality (M1 – PFC). The mean dimensionality difference is shifted negatively, highlighting lower dimensionality in M1 relative to PFC. Dashed lines: equality between PFC and M1 dimensionalities. Note the shift towards PFC, confirming higher dimensionality in PFC.

that the extracted low-dimensional dynamics are functionally relevant for context-dependent behavior.

To confirm the robustness and specificity of this correlation, we compared the observed correlation coefficient to two surrogate null distributions (Fig. 4b, right). First, we shuffled the source and target regions across participants to disrupt the communication between PFC and M1. This permutation approach yielded a new set of surrogate CSs per participant, which however, demonstrated no systematic relationship with behavior (shuffled pairings; observed correlation relative to surrogate distribution: $P = 0.0040$, $z = 3.05$). Second, we shuffled the regression weights of the identified CS within each participant, to alter the subspace structure (Methods). The resulting surrogate subspaces also exhibited no correlation with behavior (shuffled weights; observed correlation relative to surrogate distribution: $P = 0.0100$, $z = 2.10$), reinforcing the functional relevance and specificity of the identified PFC–M1 CSs in capturing behaviorally relevant interactions. In both cases, the observed behavioral correlation ($r_{CS} = 0.79$) exceeded the boundaries of the null distribution. To address the possibility that the behavioral correlation was driven by an evoked movement potential propagating from PFC to M1, we repeated the analyses after subtracting the condition-specific trial average across channels, thereby removing the evoked response. The key findings remain intact with the behavioral relevance confined to the second CS dimension (Supplementary Fig. 4).

To further examine the behavioral relevance of the CS, we tested whether this relationship held at the single-trial level within individual participants. For this, we projected the PFC HFA for each trial onto the first dimension of the identified PFC–M1 CS. We then correlated single-trial reaction times with the timing of PFC HFA peaks (orange; $r_{PFC} = 0.36$, $P_{PFC} = 0.0031$, Spearman rank-order correlation; Fig. 4c, left, for a representative participant) and with the latent dynamics' peaks in the first dimension of the CS (purple; $r_{CS} = 0.61$, $P_{CS} = 7.5 \times 10^{-6}$, Spearman rank-order correlation). Critically, latent dynamics within the PFC–M1 CS emerged as a stronger predictor of single-trial reaction times as compared to PFC HFA peaks across participants ($P = 0.0009$, $d = -1.18$, Wilcoxon signed-rank test; Fig. 4c, right). Although M1 HFA peaks also correlated with single-trial reaction times ($r_{average} = 0.81$, Spearman rank-order correlation), their predictive strength did not differ significantly from that of CS peaks at the group-level ($P = 0.1963$, Wilcoxon signed-rank test). This observation underscores the relevance of such low-dimensional subspaces in relaying behaviorally relevant information on single trials. To determine how much of this single-trial predictive power reflects reaction time-related variance originating within PFC, we defined a PFC RT-predictive subspace using a time-resolved, cross-validated regression approach (Methods). Projections onto this subspace showed significantly higher correlation with reaction time following target onset compared to baseline ($P < 0.05$ versus a

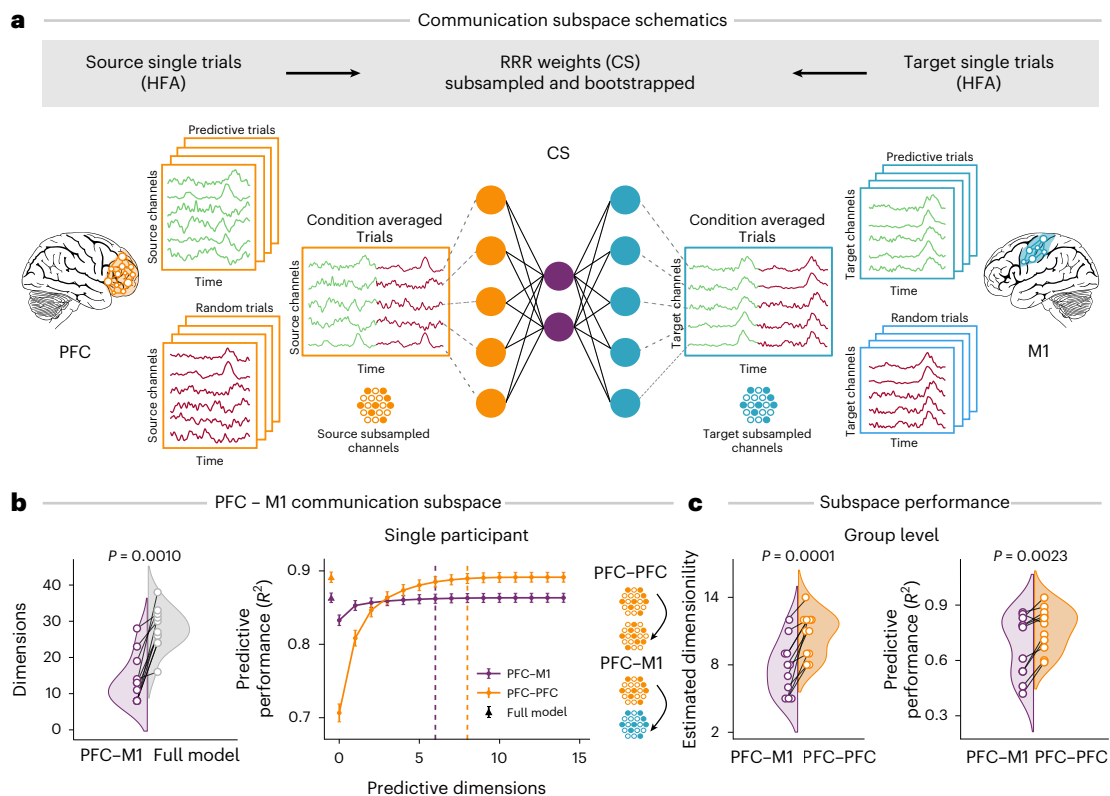


Fig. 3 | CS analysis between PFC and M1. a, Identification of a CS within PFC that is predictive of M1 activity: HFA trials from individual participants were averaged across context conditions (predictive and random) and concatenated for each recorded channel in the source (PFC, left) and target regions (M1, right). Using RRR (Methods), the dimensions of PFC population activity predictive of M1 were identified. If PFC–M1 interactions occur through a subspace, then a small subset of predictive dimensions (purple) will be sufficient to capture the interactions between the two regions. These predictive dimensions are defined by the regression weights and the dimensions required to capture PFC–M1 interactions form the CS. Dimensions that are not predictive of the M1 population activity will lie outside of this subspace. PFC channels were split into two equal sets to construct a source-to-source control subspace (PFC–PFC), where one subset of PFC channels predicted the other (within-PFC interactions). The same split was used to predict M1 activity, yielding the PFC–M1 subspace. **b**, Predicting M1 population activity from PFC population activity. Left: significantly fewer predictive dimensions (PFC–M1, purple) are required to predict the M1 population activity, while maintaining comparable predictive performance compared to using all available dimensions (full model—all recorded channels,

gray; $n = 11$; $P = 0.0010$, two-sided Wilcoxon signed-rank test). Right: predictive performance (R^2) as a function of the number of predictive dimensions of CS for a single participant. Purple and orange curves correspond to PFC–M1 and PFC–PFC subspace, respectively. Triangles indicate predictive performance of the full regression model (Methods). Curves show the mean R^2 across cross-validation folds; error bars: $\pm$ s.d. Increasing the number of predictive dimensions did not improve the predictive performance (R^2) for both PFC–PFC and PFC–M1 subspace. Dashed lines: estimated dimensionality for an example single participant, computed as the smallest number of dimensions for which R^2 was within 1 s.e. of the peak performance. Notably, fewer dimensions were required for PFC–M1 predictions compared to PFC–PFC ($n = 11$; $P = 0.0010$, $t_{10} = -5.95$, $d = -1.79$; two-sided paired t -test). **c**, PFC–M1 CS characteristics. Left: at the group-level significantly fewer dimensions were required for the PFC–M1 subspace compared to PFC–PFC ($n = 11$; $P = 0.0010$, $t_{10} = -5.95$, $d = -1.79$; two-sided paired t -test). Right: PFC–PFC subspaces achieved consistently higher predictive performance as opposed to PFC–M1 subspaces ($n = 11$; $P = 0.0023$, $t_{10} = -4.05$, $d = -1.22$; two-sided paired t -test). Circles: individual participant, with dashed lines pairing the same participant across conditions.

permutation-derived null distribution), confirming that it captured RT-relevant neural variance (Supplementary Fig. 5a). However, alignment between the reaction time-predictive subspace and the PFC–M1 CS was low at the group level (mean cosine similarity = 0.09 ± 0.01 ; Supplementary Fig. 5b). This limited overlap indicates that the CS does not simply reflect RT-driven variability within PFC. Instead, it captures a distinct component of the PFC population activity that relays behaviorally relevant information to M1.

To rule out trivial inheritance of behavioral correlation from M1, we performed ablation and temporal-permutation controls (Methods), both of which sharply reduced predictive performance, confirming that structured PFC dynamics underlie the CS (Supplementary Fig. 6). We also tested whether temporal fluctuations between PFC and M1 could account for this effect using pseudo-PFC traces, which retained temporal structure but disrupted channel covariance. Although predictive performance of the pseudo-CSs remained high, they did not correlate with context-dependent action ($P = 0.024$; Supplementary Fig. 7). Finally, comparison to targeted dimensionality reduction⁴⁵ applied

to PFC alone (Methods) showed that, although other task-related PFC subspaces exist, only the PFC–M1 CS captured the component of PFC activity relayed to M1 that is functionally relevant for guiding context-dependent action (Supplementary Fig. 8).

In sum, these results demonstrate that the identified PFC–M1 CS represented neural dynamics predictive of behavior. This approach uncovered behaviorally relevant neural activity patterns that were embedded within the high-dimensional electrophysiological recordings. Critically, the identified CS solely relied on neural activity patterns, and not on condition-labeled single trials. Hence, these findings reveal a population-level mechanism that explains how behaviorally relevant information is routed from human PFC to M1.

Discussion

Neural activity underlying human goal-directed behavior is inherently time-varying and high-dimensional. Thus, identifying the neural basis of cognitive operations and complex human behaviors remains challenging. Using direct brain recordings in humans, we demonstrate

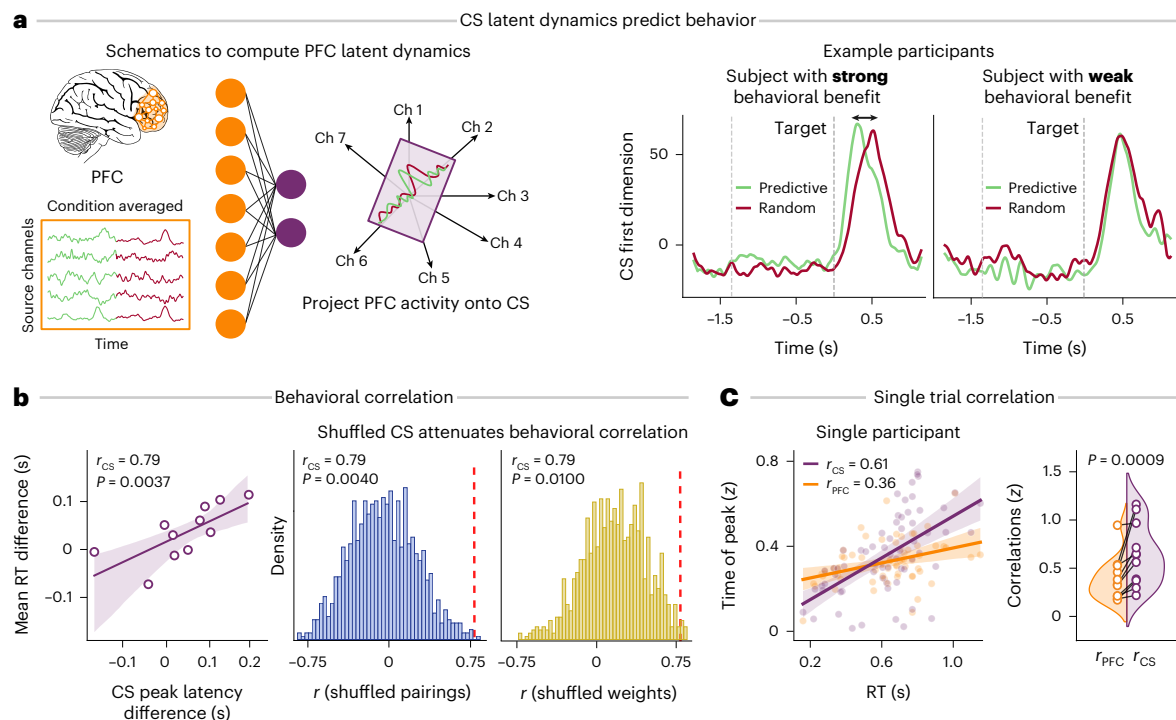


Fig. 4 | Latent dynamics of the PFC–M1 CS predict behavior. **a**, Schematics to compute PFC latent dynamics within CS. Left: PFC HFA traces are condition-averaged by concatenating trial-averaged predictive and random conditions for each channel, then projecting onto the first dimension of the CS to extract the latent dynamics. Right: latent dynamics within the PFC–M1 CS, for two example participants, show context-specific trajectories (green: predictive; red: random). Participants with strong behavioral benefit (larger mean RT difference between context-conditions) show dissociation of predictive and random target-related peaks. Participants with weak behavioral benefit (small difference in mean RT between context-conditions) show no clear dissociation of context-dependent peaks. **b**, Behavioral correlation. Left: group-level correlation between CS peak latency differences and mean RT differences, showing significant behavioral alignment ($r_{CS} = 0.79$, $P = 0.0037$; two-sided Spearman rank correlation). Circles: individual participants; shaded areas: 95% CI. Right: null distributions generated from control analyses to test the robustness of the behavioral correlation.

Null distributions are created by shuffling source and target region between participants (left distribution: permutation test, $n = 1,000$, one-tailed, $P = 0.0015$) or by shuffling the regression weights of the CS, for each participant (right distribution: permutation test, $n = 1,000$, one-tailed, $P = 0.0110$). Red dashed lines: observed behavioral correlation ($r_{CS} = 0.79$), for comparison. **c**, Single-trial correlation between peak timing and RT. Left: example from a single participant comparing the correlation between RT and the time at which the CS peaks (purple; $r_{CS} = 0.61$; two-sided Spearman rank correlation) versus the time at which HFA in PFC peaks (orange; $r_{PFC} = 0.36$; two-sided Spearman rank correlation). Each circle is a single trial. Solid lines: least-squares linear regression fit; shaded regions: 95% CI of the fitted regression line. Right: group-level comparison of Fisher z-transformed correlation values shows that CS peaks (purple) are significantly more correlated ($n = 11$; $P = 0.0009$, $d = -1.18$; two-sided Wilcoxon signed-rank test) than PFC HFA peaks (orange).

that context-dependent action information is embedded within high-dimensional PFC dynamics (Fig. 2) and is relayed to M1 for execution through a low-dimensional CS (Fig. 3). The identified PFC–M1 CS efficiently abstracts high-dimensional prefrontal dynamics into a lower dimensional representation of motor activity. Although the CS was defined on simultaneously recorded population activity from both PFC and M1, it predicted behavior on the single-participant and single-trial level (Fig. 4). Collectively, these results demonstrate how coordinated large-scale population activity encodes and relays predictive information to guide adaptive behavior.

The PFC is integral to a variety of cognitive processes and integrates sensory evidence with current rules, mnemonic information and long-term goals to coordinate behavior in a context- and demand-dependent manner^{1,46,47}. PFC activity has been shown to be high-dimensional, providing the functional basis for the large representational space needed to support these various cognitive operations^{2,9,10,12}. By contrast, reports on motor cortex (M1) dimensionality have been more heterogeneous. Earlier studies frequently described M1 dynamics as low-dimensional, consistent with its role as a primary output area optimized for efficient motor control^{17–20}, with limited contributions to complex mental operations compared to PFC^{22,48}. However, recent evidence challenges this view, suggesting that this apparent low dimensionality in M1 may arise from task constraints or

recording modalities rather than reflecting an intrinsic property of the region⁴⁹. Indeed, growing work indicates that low-dimensional structure in M1 often reflects context- or task-invariant features, whereas movement-specific variables are embedded in higher-dimensional components that explain less variance^{20,50,51}.

We approximated population dimensionality from densely sampled ECoG recordings using factor analysis⁴² and principal component analysis (PCA), which replicates and extends similar observations in the nonhuman primate motor system that were obtained primarily using similar methods^{17,18,24}. Given the relatively simple motor output (a button press) required in our visual detection task, the lower dimensionality of M1 activity might also be interpreted as reflecting the limited degrees of freedom in the motor output, rather than an inherent difference in the dimensionality of the two regions (Fig. 2b). Furthermore, dimensionality estimates were based on capturing most of the variance in neural activity, not on determining the full set of modes required to explain motor behavior. Thus, the observed dimensionality difference could reflect primarily the relative complexity of neural representations engaged by the task, with PFC encoding richer context and expectation signals, and M1 supporting a constrained motor action.

We show that a low-dimensional CS within PFC selectively transforms high-dimensional, context-dependent PFC activity into a

lower dimensional representation of the motor output (Fig. 3b,c). The results provide evidence that context-dependent action information is embedded within a restricted set of dimensions inside otherwise high-dimensional PFC dynamics. But what is the advantage of low-dimensional information coding? Low-dimensional subspace coding provides several computational benefits^{29,52}, including information separability, abstraction and generalizability, enabling efficient decoding for downstream regions²³. For example, it has been shown that several cognitive representations are represented simultaneously in distinct subspaces, thereby reducing interference^{35,53,54}. The present results demonstrate that a PFC subspace selectively communicates behaviorally relevant information to the motor cortex for action execution (Fig. 4a), hence constituting a secure communication pathway within the broader PFC activity, shielding predictive information from interference and allowing for robust decoding by M1. Mounting evidence across several cognitive domains further supports the idea that task-relevant representations are captured efficiently by a small number of dimensions^{27,28,55}. A clear example of this is the sequential unfolding of latent low-dimensional trajectories within the neural population activity space that characterize action execution^{17,19,20}, especially in the nonhuman primate motor system. However, it is critical to remember that these low-dimensional dynamics exist in a much larger, more complex high-dimensional space, even in the motor cortex⁴⁹. This higher dimensional space has been shown to house latent representations for alternative or intended actions that are currently inactive⁵⁶. Our present work extends these observations by identifying a behaviorally relevant subspace within the high-dimensional space of complex PFC neural activity.

We approached the question of context-dependent target detection from the perspective of two communicating populations, as previous lesion and scalp electroencephalogram evidence have implicated the PFC in contextual processing^{40,57}. Our results revealed that activity relayed from human PFC to M1 can be captured within a specific latent subspace. Critically, the identified subspace relayed information about the ensuing behavior on an individual trial-by-trial basis (Fig. 4b,c), surpassing the modest correlations between grand-average HFA responses and behavior (Fig. 2a). These observations are in line with the distributed nature of coding in the PFC and further suggest that task-relevant information may be obscured in grand-averages by incorporating representations that encode both task-relevant and task-irrelevant information^{1,12,16}. It is essential to highlight that the CS was defined solely on neurophysiological criteria based on aggregated mean HFA, thus, the subspace model neither incorporated single-trial dynamics nor had access to behavioral information. Nevertheless, its activity correlated with ensuing behavior on the participant and trial level (Fig. 4b,c) and explained inter-person differences on how participants incorporated predictive information into the target detection process (Fig. 4a,b). Moreover, activity in the CS emerged only after target presentation (Fig. 4a), highlighting its temporal specificity. It is important to note that other functionally relevant low-dimensional subspaces exist within PFC (Supplementary Fig. 8). However, among the subspaces we evaluated, the PFC–M1 CS transmitted action information from PFC to M1 most effectively. This observation aligns with previous work highlighting the flexibility of PFC subspace coding that multiplexes information across different functional subspaces to support complex cognitive processes collectively⁵⁸. Although the build-up of choice predictive activity in the prefrontal motor network might emerge even before the onset of predicted targets, this activity was not captured by the most predictive dimension of the CS. Given the sparse spatial sampling that did not include sensory regions, we cannot fully resolve where choice predictive information is integrated. Note that clinical considerations dictated electrode placement; hence, visual areas were not explored in the study participants. Nevertheless, previous studies in both humans and mice demonstrated that early sensory regions encode context-dependent information^{54,59,60}. The absence of similar context-dependent effects in

M1, together with previous causal evidence linking PFC to the biasing of motor output using predictive information in a highly similar visual target detection task⁴⁰, suggests that contextual modulation originates in PFC and becomes visible through changes in the timing of activity peaks. Together, these findings support the view that PFC provides the contextual drive that biases motor preparation through an interareal CS.

Although coding subspaces are typically defined on aggregated population spiking activity^{28,29}, we studied HFA in humans, which serves as a viable surrogate of coordinated population activity. Previous work has shown that HFA correlates with latent dynamics in specific brain regions⁶¹, as well as with underlying spiking activity^{37,39,62}. This suggests that both HFA and spiking activity constitute viable abstractions of the same underlying population dynamics^{61,62}. In summary, these results are in accord with previous evidence in humans and nonhuman primates that demonstrated that behaviorally relevant information can be extracted from high-dimensional neural activity patterns and extend it to show how differences in neural dimensionalities reflect the respective roles of the PFC and M1 for action planning and execution.

Interareal communication has been studied mostly using the functional connectivity framework, that is, correlation- or coherence-based bivariate approaches that consider interaction between two recording sites^{63,64}. However, it is unclear how this framework relates to distributed population-based coding, especially when the sender (PFC) is higher dimensional than the receiver (M1). Although theoretical frameworks are well developed²³, experimental evidence remains limited. One recent human intracranial electroencephalogram study bridged population- and synchrony-based modes of cortical communication by demonstrating that the subspace that relayed action information from PFC to M1 exhibited a high-degree of oscillatory theta synchrony³⁴. Almost all the experimental evidence for CSs comes from studies in nonhuman primates^{24–26,30} and mice³¹. Whereas many of these studies focused on the underlying neurophysiology^{24,31}, more recent work in nonhuman primates has shown that CSs flexibly route interareal information during decision-making and attention^{25,30}. Here we reveal the presence of a CS between PFC and M1, providing evidence that similar principles govern higher-order cognitive operations in humans. We find that context-dependent action information in the first dimension of the CS correlated significantly with the behavioral benefit afforded by predictive information on visual target detection (Fig. 4b). Although individual participants may exhibit additional predictive dimensions (Fig. 3c), the first dimension provided a stable reference for group-level analyses, allowing us to identify behaviorally relevant signals despite interindividual variability (Supplementary Fig. 9). Additional dimensions probably capture other aspects of interareal communication but contribute less to explaining the contextually driven motor output, suggesting that behaviorally relevant information is routed preferentially along specific low-dimensional axes. This organization may reflect a general coding principle by which high-dimensional representations in higher-order cortical regions are distilled into the behaviorally most salient components before reaching downstream targets, allowing for efficient and flexible control of action. In addition, the PFC–M1 CS was defined using condition-averaged activity, which emphasizes task-evoked covariance between PFC and M1 rather than moment-to-moment fluctuations, in contrast to previous work in nonhuman primates²⁴. This distinction was evident when we computed the subspace using baseline activity. The resulting subspace no longer predicted context-dependent actions (Supplementary Fig. 10). These results indicate that the PFC–M1 mapping reconfigures as task demands evolve. Specifically, we hypothesize that the behaviorally relevant subspace mediates primarily the relay of context-dependent action signals from PFC to M1 rather than reflecting predictive sequence processing in general, which may be less directly coupled to motor output. Future work could examine how the stability of CSs is shaped by the temporal dynamics of neural activity within different regions, and how this stability affects the routing of behaviorally relevant information.

Notably, we employed broad definitions of PFC and M1 given that we recorded from clinical macroelectrodes that covered larger portions of PFC. Recording single-unit activity from dorsolateral PFC and M1 is not feasible using conventional microwires, requiring, for example, microarray or intraoperative neuropixel recordings⁶⁵ that are not performed routinely. Because the task was designed as an RT paradigm with a clearly defined suprathreshold target, participants performed at near-ceiling accuracy⁴⁰, leaving too few error trials for meaningful analysis. The cohort size was also modest ($n = 11$), reflecting the rarity of simultaneous PFC and M1 intracranial coverage, which limited group-level comparisons across conditions. To mitigate any potential bias from the sparse sampling of subdural grid electrodes, we bootstrapped the bipolar pairs to match contacts between regions. In sum, our results replicate and extend previous observations in the nonhuman primate visual system and establish the feasibility to study interareal communication in the human brain through the lens of a population-coding framework.

Conclusions

Collectively, our results reveal a fundamental coding principle that supports context-dependent action execution in humans. The identification of a behaviorally relevant PFC–M1 CS opens new avenues to conceptualize and study coordinated interareal population activity that enables cognitive control.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-026-02290-4>.

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Methods

Participants

We obtained ECoG recordings from a total of 12 pharmacoresistant individuals with epilepsy (age 40.91 ± 12.32 years, mean $\pm$ s.d.; four female individuals, one unknown) who underwent presurgical monitoring and were implanted with subdural grid electrodes. Data from one individual was excluded from all neural analyses, except for behavioral analyses, owing to high-frequency artifacts, which precluded analyses of HFA. Electrode implantation sites were determined exclusively based on clinical needs. All procedures were approved by the Institutional Review Boards of University of California (UC) Berkeley (Berkeley Committee for Protection of Human Subjects Protocol no. 2010–02–783) as well as the Institutional Review Boards at the clinical sites at UC San Francisco (UCSF), UC Irvine, Stanford School of Medicine and Johns Hopkins University School of Medicine. All participants provided written informed consent.

ECoG data acquisition

ECoG data were recorded using different acquisition systems. At the UCSF and Stanford, data were collected with a 128-channel recording system (Tucker-Davis Technologies), typically at 3,052 Hz. At Johns Hopkins, data were recorded at 1,000 Hz using a 128-channel system (Stellate Systems). At the UC Irvine Medical Center, recordings were performed using a 256-channel Nihon Kohden system (model JE120A). Signals underwent analog filtering at 0.01 Hz and were sampled digitally at 5,000 Hz. ECoG data were recorded using a subdural electrode as reference (an electrode with minimal or stable signal located away from cortical areas of interest) and a scalp electrode as ground. Sampling rates, online filters and amplification across all recording systems were set to allow comparability across recording sites for HFA. As outlined below, we re-referenced all recordings to adjacent bipolar pairs, extracted HFA, which was further z-normalized to mitigate any impact of the recording system.

Electrode localization

A structural preoperative magnetic resonance imaging (MRI) scan as well as a postimplantation computed tomography (CT) scan was acquired for all participants. The MRI and CT scans were coregistered to the same space using two nonlinear transformations based on normalized mutual information implemented in the Bioimage suite⁶⁶ or in the FieldTrip toolbox⁶⁷ as described previously⁶⁸. In brief, we transformed both the pre-implant MRI and the postimplant CT into Talairach space. Then we segmented the MRI using Freesurfer⁶⁹ and then coregistered the T1 to the CT. 3D electrode coordinates were determined on the coregistered CT scan. We corrected for the brain shift that is often observed after large craniotomies⁷⁰. Finally, we warped the aligned electrodes onto a template brain in Montreal Neurological Institute (MNI) space. For all electrodes, we assigned a label from the Analysis of Functional Neuroimages atlas⁷¹. We verified and corrected the assigned atlas position manually after inspection of the electrode reconstruction in native Talairach space.

Experimental procedure

Participants performed a context-guided visual target detection task^{40,41,72} (Fig. 1a), in which they viewed a continuous stimulus train consisting of flashing triangles facing left, up or right, presented against a gray background. The task contained two context conditions: predictive, where T was preceded by a predictive sequence of stimuli (S1, S2, S3 or left, up, right-facing triangles) and random, where T appeared randomly within the stimulus stream. Participants were informed that the predictive sequence would always (100%) be followed by T and were instructed to respond with a button press as quickly as possible, once the target appeared on the screen. The two trial types (predictive and random targets) were intermixed within blocks and fully randomized across trials. Each participant completed four to five blocks, with equal

numbers of predictive and random targets in each block. Trials were epoched to include the three stimuli preceding the target, with every epoch event locked to the onset of T. For all subsequent analyses, a time interval of interest (TOI) was defined, spanning from 500 ms before the onset of the final stimulus in the sequence to 1.15 s after the onset of T. This TOI was selected to capture both the preparatory and response phases of the task, allowing for a detailed examination of how context guides target detection.

Regions of interest

All artifact-free electrodes that covered the precentral gyrus were classified as motor cortex (M1) electrodes. All electrode that covered any prefrontal area, including lateral, medial or orbitofrontal cortex, were classified as PFC according to the atlas-based detections. All electrode assignments were confirmed visually and, if necessary, corrected manually.

Behavioral analysis

We analyzed RTs from participants performing the context-guided target detection task. A paired statistical test was performed to evaluate differences in RTs between predictive and random contexts. Based on normality assumptions, we either used a paired *t*-test if the both groups were normally distributed or Wilcoxon signed-rank test, if either of the groups did not follow a normal distribution. The Shapiro–Wilk was used to assess whether the groups were distributed normally.

ECoG preprocessing

ECoG signals were preprocessed by demeaning, linear detrending, and applying a local bipolar referencing scheme (referencing each channel to its nearest neighbors on the grid). Given that most grids featured an 8×8 layout, all electrodes exhibited several neighbors, and we included all unique combinations along horizontal, vertical and diagonal orientations. Line noise and its harmonics were removed using a notch filter at 60 Hz. A board-certified neurologist performed a visual inspection of the raw data to identify and remove any epileptic activity or artifacts. Channels and epochs containing interictal epileptic discharges or other artifacts were excluded from further analyses.

HFA extraction

HFA, also referred to as high gamma or high-frequency broadband activity, is the broadband activity spanning the 70–150 Hz range and is measurable exclusively using intracranial recordings owing to the low-pass filtering properties of the skull. HFA has been established as a reliable marker of human behavior, offering high spatiotemporal resolution^{15,21,39,73}. It serves as a surrogate for multi-unit activity in humans⁷⁴, but recent evidence also suggests that HFA contains additional behaviorally relevant information^{34,62}. This renders HFA a particularly valuable metric for characterizing population-level neural activity, enabling the study of large-scale cortical networks in humans. To extract the HFA timeseries, we first bandpass-filtered the cleaned data epochs into seven nonoverlapping 10 Hz wide bins (± 5 Hz) with center bins spaced linearly between 75 Hz and 145 Hz range. The instantaneous amplitude of each filtered timeseries was then computed using the Hilbert transform⁷⁵. Next, high-frequency traces were normalized by constructing a bootstrapped baseline distribution. This was achieved by resampling baseline values (from -4.5 to -4.0 s, relative to the onset of the first stimulus in the sequence) randomly 1,000 times with replacement. Each high-frequency trace was normalized by subtracting the mean of the bootstrap distribution and dividing by its s.d. The final HFA timeseries was obtained by averaging the seven individual high-frequency traces, yielding a single representation of HFA.

HFA peak latency difference correlation

HFA peak latency differences between predictive and random contexts were analyzed in PFC and M1 and correlated subsequently with mean

RT differences. Individual mean RT differences were computed as the difference between the mean RT across all random trials and the mean RT across all predictive trials. For each participant, condition-specific HFA traces were averaged across trials and channels. HFA peaks were defined as the maximum HFA amplitude in the time window between target onset and the slowest single-trial RT to ensure that the identified peaks were related temporally to the behavioral responses. Subsequently, the latency difference of the peaks between contexts was quantified by subtracting the timepoints corresponding to the peaks in the predictive condition from that in the random condition. To evaluate the association between neural and behavioral measures, the nonparametric Spearman rank-order correlation coefficient was computed between the individual mean RT differences and HFA peak latency differences, yielding a group-level correlation coefficient. The significance (P value) Spearman's rank correlation was assessed using a t -distribution.

Factor analysis

To quantify the dimensionality of neural activity in the PFC and M1 populations for every individual participant, we used factor analysis⁴². Factor analysis models the observed high-dimensional population neural activity (y) as arising from a smaller set of latent factors (z), using the relationship:

$$\begin{aligned} z &\sim \mathcal{N}(0, I) \\ y|z &\sim \mathcal{N}(Lz + \mu, \Psi) \end{aligned}$$

where y is a q -dimensional vector representing the observed neural activity, with q denoting the number of observed variables (for example, the number of recorded channels). The variable y is conditioned on z and $\mathcal{N}$ represents the multivariate normal distribution. The latent factors z are m -dimensional, where $m < q$ and their linear relationship with the observed data y is described by the loading matrix L , which has dimensions $q \times m$. L maps the m -dimensional latent factors onto the observed q -dimensional activity. μ is the mean vector associated with y , and Ψ is a $q \times q$ matrix representing the unique variances (specific noise) of each observed variable. These equations encapsulate the generative model for factor analysis, where the latent factors z drive the observed variables y , with noise specified by Ψ . To estimate the dimensionality of the latent factors z driving the population activity, we followed the same two-step procedure as described recently²⁴. First, we found the number of dimensions m_{peak} that maximized the cross-validated log-likelihood of the observed data through a grid search. In the second step, we fit a factor analysis model with m_{peak} dimensions, and the dimensionality m was defined as the smallest number of latent factors required to capture 95% of the variance in the shared covariance matrix (LL^T), such that the cumulative variance reflects how well successive latent dimensions account for the structure captured in the shared covariance. As a result, the cumulative curve is normalized to the total shared variance, without indicating that all variance in the raw neural activity is explained.

This analysis was conducted on a trial-by-trial basis for each participant, by fitting the factor analysis model to simultaneously recorded PFC and M1 population activity on single trials. For each trial, the model yielded an estimate of the latent dimensionality, enabling a robust estimation of the dimensionality of individual brain regions. Furthermore, the variance explained by each dimension was computed using the eigenvalues (λ) of the shared covariance matrix (LL^T). For group-level comparisons, the final dimensionality for each participant was estimated as the average cross-validated dimensionality across all trials. This approach provided a quantification of the number of latent factors capturing shared variance in neural population activity, facilitating group comparisons of population complexity between the PFC and M1 regions.

Participation ratio

To further quantify differences in the dimensionality of population activity between PFC and M1, we employed the participation ratio (PR) measure⁷⁶. The PR is derived from the eigenvalues (λ) of the neural covariance matrix with dimensions N , which forms the basis of factor analysis or PCA and reflects the number of latent variables or factors required to account for the variance in the observed data. The PR is calculated using the formula:

$$\text{PR} = \frac{(\sum_{i=1}^N \lambda_i)^2}{\sum_{i=1}^N (\lambda_i)^2}$$

This metric provides an alternative characterization of dimensionality by quantifying the effective number of dimensions contributing to the population activity, with higher values indicating greater dimensionality.

Subsampling and regression performance

To assess action information encoded in the PFC, we used ridge regression to estimate how well fluctuations in M1 population activity could be predicted from fluctuations in PFC population activity, while controlling for unequal sampling of recorded channels. This approach is referred to as the full regression model throughout the text. For each participant, HFA activity recorded from PFC was represented as a tensor of shape $[tr \times p \times t]$, and from M1 as $[tr \times k \times t]$, where tr denotes the total number of trials, p , k represent the number of recorded channels in PFC and M1, respectively, and t corresponds to the timepoints within the TOI. Trials were segregated by context ($i = 2$; predictive or random) and averaged within each condition, resulting in activity matrices of shape $[it \times p]$ for PFC and $[it \times k]$ for M1. To ensure consistency across participants, we constrained the analysis to the smallest number of available cleaned recording channels in either region ($n = 16$), setting the degrees of freedom to $l = 16 - 1 = 15$. Before subsampling, the PFC population activity was further divided into two subsets of shape $[it \times (p/2)]$ to generate PFC–PFC and PFC–M1 subspaces. Subsampling was performed by randomly selecting l channels with replacement from each subset of PFC and M1, repeated over 1,000 iterations per participant to generate surrogate distributions.

Ridge regression was conducted in the form $Y = XB + \epsilon$, where Y (shape; $[it \times l]$) represents M1 activity, X (shape; $[it \times l]$) represents PFC activity, B (shape; $[l \times l]$) denotes the weight matrix mapping predictors to targets and ϵ represents the residual error. The weight matrix B was estimated by minimizing the regularized loss function $\mathcal{L}(B) = \|Y - XB\|_F^2 + \alpha \|B\|_F^2$, where $\|\cdot\|_F$ is the Frobenius norm⁵⁶, representing the sum of squared errors and α is the regularization parameter. The solution for B was computed as $B = (X^T X + \alpha I)^{-1} X^T Y$, where I is the identity matrix. The optimal α for each participant was determined by tenfold cross-validation, defined as the largest value yielding performance within 1 s.e. of the best-performing model. The optimal α values were stored for further analyses. The full regression model prediction performance (R^2), serving as a benchmark, was calculated as the mean bootstrapped prediction accuracy, estimated across a tenfold cross-validation using the determined optimal α . This robust procedure quantified the relationship between PFC and M1 activity, ensuring reliable comparisons across participants.

Regression models

To test the existence of a low-dimensional CS^{23,24} between the source (PFC) and target (M1) population activity, we employed RRR, as described in previous studies^{24,43}. Specifically, we tested whether M1 population activity could be predicted by a subspace of PFC population activity as well as by the full regression model, which used all recorded channels. The RRR solution is defined as:

$$B_{\text{RRR}} = B_{\text{OLS}} V V^T$$

Here, $B_{OLS} = (X^T X)^{-1} X^T Y$ represents the ordinary least-squares (OLS) solution, which minimizes the squared prediction error. The matrix V , of size $l \times m$, defines a restricted subspace of rank m , where m is the dimensionality constraint imposed on the regression model. The columns of V are the top m principal components of the OLS prediction, $\hat{Y}_{OLS} = X B_{OLS}$, ordered by decreasing eigenvalues (from the most explained variance to the least). To predict the M1 target population activity using RRR, we computed:

$$\hat{Y}_{RRR} = X B_{RRR} = X B_{OLS} V V^T = X \bar{B} V^T$$

Here the matrix $\bar{B} = B_{OLS} V$, of size $l \times m$, contains the dimensions of the source PFC population activity that are used for predicting M1. These dimensions are referred to as predictive dimensions in the text. The rank of the model is defined as $\text{rank}(\bar{B}) = m$. To determine the estimated dimensionality of the CS (m), we once again applied a bootstrapped tenfold cross-validation procedure, where the model was trained on 90% of the condition-averaged data and tested on the remaining 10% in each fold. The estimated dimensionality was defined as the smallest number of dimensions for which the predictive performance (R^2), was within 1 s.e. of the peak performance. Final R^2 values reported in Fig. 3b,c therefore reflect prediction accuracy on held-out data, averaged across all folds and bootstrap permutations. This method enabled us to isolate the dimensions of PFC activity that are predictive of the motor output, and the top predictive dimensions were deemed as the PFC–M1 CS.

M1–M1 control analysis

To verify that the low dimensionality of the PFC–M1 subspace was not a trivial consequence of the intrinsic dimensionality of M1, we performed a control analysis comparing PFC–M1 with matched within-M1 interactions. One participant with insufficient M1 coverage (S08; 16 electrodes) was excluded, and the number of subsampled channels per iteration was reduced from 15 to 10 to ensure nonoverlapping M1 populations (owing to the reduced electrode coverage in M1). Subsampling and ridge regression followed the same procedure as the PFC–PFC analysis, with predictor and target sets drawn from the same M1 population. Dimensionality and predictive performance were computed across iterations. By holding the M1 target population fixed, this analysis ensured that any observed differences in dimensionality reflect selective projections from PFC rather than limitations of the target population.

Source projection and latent dynamics

To extract context-dependent action information embedded within high-dimensional source PFC population activity, we projected it onto the PFC–M1 CS obtained through RRR, as outlined above⁴⁴. This projection highlights the low-dimensional structure of activity that contributes to action information predictive of the motor output. The context-condition-averaged source activity X was mean-centered to remove biases and the resulting centered activity $\tilde{X}$ was projected onto the subspace defined by the RRR weights B_{RRR} to compute latent dynamics, using the following relation:

$$X_{\text{proj}} = \tilde{X} B_{RRR}$$

resulting in the projected matrix X_{proj} with shape $[it \times m]$, where m is the estimated dimensionality of CS for individual participants, computed in the previous step. This matrix represents the latent dynamics of the source PFC population activity in the PFC–M1 CS. For each participant, the extracted latent dynamics X_{proj} yields condition-specific latent dynamics that can be studied separately. Hence, the projection preserves the trial-specific temporal evolution of context-dependent neural activity while aligning it with the subspace.

Latent dynamics peak latency correlation

Peak latency differences of latent PFC dynamics between predictive and random contexts were analyzed in the PFC–M1 CS and correlated subsequently with mean RT differences. For each participant, latent dynamics in the predictive and random contexts were derived in a time window between target onset and the slowest single-trial RT. The peaks of the latent dynamics were defined as the maximum amplitude in the first dimension of the subspace, as this dimension captures the most explained variance in the activity and is the most predictive of M1. The latency difference of these peaks between the predictive and random contexts was computed by subtracting the peak time point in the predictive context from that in the random context. Finally, Spearman rank-order correlation coefficient was computed between individual mean RT differences and latent dynamics peak latency differences, yielding a group-level correlation coefficient and the significance (P value) Spearman's rank correlation was assessed using a t -distribution.

Null distributions to test behavioral correlation robustness and subspace specificity

To assess both the robustness of the behavioral correlations and their specificity to the identified subspace, we constructed two null distributions. We tested the significance of the observed correlation between the first component of the PFC–M1 CS and context-dependent behavior against two null hypotheses:

- (1) Weight-shuffle null: we disrupt the structure of the identified subspace by randomly permuting the subspace weights ($n = 1,000$ permutations).
- (2) Cross-participant null: we form 'mismatched' CSs by pairing PFC data from participant A with M1 data from participant B ($A \neq B$), generating 1,000 such permutations per pairing.

In both cases, these procedures remove any true relationship between PFC and M1, allowing us to estimate a null distribution of correlations under the assumption of no meaningful interareal communication. The P values we report are from one-tailed, nonparametric Monte Carlo permutation tests, defined as the proportion of shuffled correlations greater than or equal to the observed value (testing specifically for positive effects). Furthermore, as the null distributions are approximately normal, we report z values in the text as a standardized measure of effect size (that is, the number of s.d. between the observed correlation and the mean of the null distribution).

Alignment between PFC RT subspace and PFC–M1 CS

To examine how much of RT-driven variance in PFC is communicated through the PFC–M1 CS, we first identified population activity patterns in PFC that predicted trial-to-trial variation in RT. For this, we constructed a time-resolved RT subspace using a cross-validated ridge regression framework. For each participant, single-trial PFC activity was segmented into nonoverlapping 50-ms bins, and neural responses were averaged within each bin to yield a trial-by-channel feature matrix. Within each bin, we fit a ridge regression model that mapped PFC population activity onto single-trial RTs, using nested fivefold cross-validation and z scoring based on training data only. The optimal regularization parameter for the ridge regression model (α), for each time bin, was determined by tenfold cross-validation, defined as the largest value yielding performance within 1 s.e. of the best-performing model. The resulting regression weights defined an RT-predictive axis for each time bin, forming a time-resolved RT subspace. To validate this subspace, held-out trials were projected onto the RT axis from each bin and the Spearman correlation between projections and RT was computed and averaged across folds, yielding a participant-specific time course of RT-predictive strength. Alignment between the PFC RT subspace and the PFC–M1 CS was quantified using cosine similarity between the RT-predictive axis for each time bin and

the first dimension of the PFC–M1 CS, which exhibited significant correlation with RT. Absolute values were used, so that all cosine similarity scores lie between 0 (orthogonal) and 1 (aligned). For each participant, the cosine similarity values reported, correspond to the averaged similarity across time bins post target onset.

CS ablation and permutation controls

To determine the importance of specific CS dimensions, we removed (set to zero) the leading dimensions of the identified PFC–M1 CS in the test data sequentially and recalculated the predictive performance (R^2) using fivefold cross-validation. Two permutation controls were implemented: (1) full permutation, in which PFC activity values were shuffled randomly across condition-averaged and timepoints and (2) block permutation, in which PFC was segmented into 100-ms blocks and then permuted randomly, preserving local temporal correlations but disrupting longer-range temporal structure. Both permutation datasets were subjected to the same ablation and cross-validation procedures. Group-level performance was calculated by averaging across participant, and s.e. values were estimated.

Targeted dimensionality reduction of PFC activity with demixed PCA

To benchmark the behavioral relevance of the identified PFC–M1 CS against other functionally interpretable subspaces within PFC, we applied demixed PCA (dPCA)⁴⁵. dPCA is a supervised dimensionality reduction method that separates population variance into components associated with distinct task parameters (such as context conditions: predictive versus random). For each individual, PFC trial data separated based on context were decomposed into two dPCA components. First, a condition-independent subspace that captured variance common to both predictive and random trials, reflecting task-invariant dynamics. Second, a context-related subspace which isolated variance that differentiates between context conditions (predictive and random). The dPCA components were defined on the same TOI as the PFC–M1 CS, spanning from 500 ms before the onset of the final stimulus in the sequence to 1.15 s after the target onset. Once the decoder weights for each subspace were obtained, we projected condition-averaged PFC activity onto the first component to extract latent trajectories. At the group level, the context-dependent difference in peak latency of this component was correlated with the corresponding difference in RTs. For single-trial analyses, individual PFC trials were projected onto the same dPCA-derived axes, and trial-specific peak latencies were correlated with RTs.

Quantification and statistical analysis

To assess statistical significance of group-level differences between two conditions (predictive versus random), we used paired statistical tests, depending on the underlying distribution of the data, unless stated otherwise. The analysis was performed on paired datapoints, such as metrics derived from latent dynamics, RTs or other relevant quantities for each participant. First, we evaluated the normality of the data for each condition using the Shapiro–Wilk test, to account for small sample size ($n = 11$ of 12). If both conditions were distributed normally (that is, P values from the Shapiro–Wilk test exceeded the alpha level of 0.05), we applied a paired t -test to compare the means of the two conditions. If either condition violated the normality assumption (that is, the Shapiro–Wilk test returned a P value below the alpha level of 0.05), we used the nonparametric Wilcoxon signed-rank test instead. In both cases, a significance threshold of $\alpha = 0.05$ was used to determine whether the null hypothesis could be rejected. A P value below this threshold indicates a statistically significant difference between the conditions, whereas a P value above the threshold suggests no significant difference.

Permutation-based cluster analysis was performed on all traces over time (Figs. 2a and 4a,b). A permutation distribution was generated

by shuffling participant labels randomly and recomputing the cluster statistic, which quantifies the strength of the observed effect across time. The permutation P value was calculated by comparing the cluster statistic to the distribution generated by random permutations. Clusters were considered significant if $P < 0.05$.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper. Additional data is available from the corresponding author upon request.

Code availability

Minimal example codes with simulated data have been made publicly available upon publication on GitHub (<https://github.com/nehabinish/pfc-m1-communication-subspace>)⁷⁷. Freely available software and algorithms used for analysis are listed where applicable. MATLAB codes for RRR and factor analysis has been made publicly available at <https://users.ece.cmu.edu/~byronyu/software.shtml> or can be accessed at <https://github.com/joao-semedo/communication-subspace.git>. Python implementations of the same and all code related to the analyses of the manuscript will be made publicly available upon publication on GitHub (<https://github.com/nehabinish/pfc-m1-communication-subspace>)⁷⁷.

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Author contributions

Conceptualization: N.B., R.F.H.; Methodology: N.B., J.T., J.M., R.F.H.; Formal analysis: N.B.; Investigation: J.J.L., R.T.K., R.F.H.; Resources: R.T.K., R.F.H.; Writing—original draft: N.B., R.F.H.; Writing—review and editing: J.T., J.M., J.J.L., R.T.K.; Supervision: R.F.H.; Funding acquisition: R.T.K., R.F.H.

Competing interests

J.M. is currently employed by Springer Nature. However, this employment began after the completion of his contribution to this research and did not influence the study design, data collection, analysis, manuscript preparation or the decision to submit the article for publication. The other authors declare no competing interests.

Additional information

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Software and code

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Data collection	Intracranial EEG data was recorded at various clinical sites using the clinical recording system and software and subsequently imported in Matlab and Python. The behavioral paradigm was implemented in Matlab, responses were collected using an analog response box that recorded as an analog channel during EEG acquisition.
Data analysis	Data analysis was carried out in Matlab and Python. MATLAB codes for reduced-rank regression and factor analysis has been made publicly available (https://users.ece.cmu.edu/~byronyu/software.shtml) or can be accessed at (https://github.com/joao-semedo/communication-subspace.git). Python implementations of the same code and code related to the analyses of the manuscript is available GitHub (https://github.com/nehabinish/pfc-m1-communication-subspace).

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Source Data files corresponding to all graphs presented in the Figures and Extended Data are provided with the manuscript. Analysis code and associated data are available on GitHub (<https://github.com/nehabinish/pfc-m1-communication-subspace>).

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Reporting on sex and gender	We recruited 12 patients who were implanted with subdural grid electrodes, which included 4 female, 1 unknown and 7 males.
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	All patients suffered from pharmacoresistant epilepsy. EEG channels exhibiting epileptic or artifactual activity were excluded prior to analysis. All participants performed the task and were able to follow the task instructions.
Recruitment	All participants were recruited during presurgical monitoring with subdural grid electrodes from UC San Francisco, UC Irvine, Stanford School of Medicine, and Johns Hopkins University School of Medicine.
Ethics oversight	All procedures were approved by the Institutional Review Boards of University of California (UC) Berkeley (Berkeley Committee for Protection of Human Subjects Protocol No. 2010-02-783) as well as the Institutional Review Boards at the clinical sites at UC San Francisco, UC Irvine, Stanford School of Medicine, and Johns Hopkins University School of Medicine.

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Sample size	We included 12 patients. We did not perform a sample size calculation, but patients were recruited based on electrode coverage. This sample size is comparable to other studies in the field (e.g., Haller et al., 2018, Nature Human Behavior).
Data exclusions	We analyzed behavioral data for all 12 patients. One patient had to be excluded from all EEG analyses due to excessive high-frequency noise that impeded subsequent analyses.
Replication	No attempt at replication at the group level was made. At the subject level, the key behavioral and EEG effects generalized across subjects.
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